

Automated Essay Scoring of College Application Essays with LLMs

Jinwoo Kim
jinwoo.kim@yale.edu

Advisor: Stephen Slade
stephen.slade@yale.edu

*A Senior Thesis as a partial fulfillment of requirements
for the Bachelor of Science in Computer Science*

Department of Computer Science
Yale University
Apr 21, 2025

Acknowledgements

This work would not have been possible without the generous support of ElevatEd, the college consulting firm that provided the essay dataset used in this study. Their willingness to collaborate made this project both feasible and meaningful. I express my sincere gratitude to Professor Stephen Slade for his guidance and thoughtful mentorship throughout the course of this research. Finally, I would like to give a special thanks to the Yale Department of Computer Science for providing a rigorous, collaborative, and inspiring environment for academic exploration.

Contents

1	Introduction	5
2	Background	6
2.1	Related Work	6
2.2	Limitations with AES and LLMs	6
2.3	Novelties	6
3	Methodology	8
3.1	Data Collection	8
3.2	Data Preprocessing	8
3.3	Prompt Engineering	9
3.4	Data Training	9
4	Results	10
4.1	Holistic Score Prediction	10
4.2	Impact of Low Trait Scores on Overall Essay Rating	11
5	Conclusions	13
6	Future Work	14

Automated Essay Scoring of College Application Essays with LLMs

Jinwoo Kim

Abstract

This study explores the application of Large Language Models (LLMs) for automated essay scoring (AES) in the context of college admissions, where evaluation extends beyond grammar and coherence to include personal voice, narrative structure, and emotional resonance. We develop a domain-specific evaluation framework inspired by professional admissions consulting practices, scoring 1,177 Common App essays across five dimensions: Problem/Solution/Takeaway (PST) Ratio, Specificity, Pace, VSPICE (a composite of six character traits), and Creativity. Using GPT-4o and rubric-informed prompts, we extract trait scores that reflect the narrative and stylistic strengths of each essay. These scores are combined with dense semantic representations generated by a sentence-level BERT encoder to form a hybrid feature vector. We evaluate multiple modeling pipelines, ranging from trait-only regressors to hybrid models incorporating both handcrafted features and transformer embeddings. Our final architecture employs a random forest regressor trained on this hybrid vector to predict human-assigned holistic scores. In addition to measuring model performance, we analyze the relationship between individual trait scores and overall ratings. We find that all five traits positively correlate with the holistic score and that essays receiving a low score in any single trait tend to exhibit both a lower median and a lower ceiling in overall scores. Although our results highlight key structural and stylistic factors that influence essay quality, they also underscore the complexity of modeling human judgment in high-stakes and narrative-driven contexts. Rather than replacing human reviewers, we envision this approach as a tool to provide formative feedback and support students through an iterative revision process. By making AES systems more interpretable and aligned with expert evaluation criteria, this work offers a foundation for future research at the intersection of education and natural language processing.

1 Introduction

In college admissions, the personal essay remains one of the few opportunities for applicants to express individuality, context, and personality beyond quantitative metrics such as GPA and test scores. However, evaluating these essays is inherently subjective, labor intensive, and costly, particularly for private admission consulting firms that must offer detailed, high-quality feedback at scale. Recent advances in Large Language Models (LLMs) have renewed interest in Automated Essay Scoring (AES) systems that can emulate or assist human evaluators. However, most existing AES systems are trained on academic writing or standardized tests, and fail to capture the narrative depth and emotional nuance central to college application essays.

This paper investigates whether LLM-based systems can provide accurate and interpretable holistic scoring of Common App personal statements. Specifically, we ask: *Can we model expert judgment of college essays using structured trait scores and transformer-based textual representations?*

While prior work has demonstrated the promise of LLMs in AES, few studies focus on narrative-driven admissions writing. Our work bridges this gap by integrating domain-specific rubrics from professional consulting into the scoring pipeline, producing interpretable intermediate metrics that reflect the values typically emphasized in admissions feedback.

Our primary contribution is the design and evaluation of a hybrid AES system that combines five interpretable, rubric-aligned trait scores—PST Ratio, Specificity, Pace, VSPICE, and Creativity—with semantic embeddings derived from BERT-based models. We employ OpenAI’s GPT-4o to assign trait scores through rubric-conditioned prompting and train several regressors to predict human-assigned holistic ratings using both structured and unstructured data.

We conduct our study on a curated dataset of 1,177 Common App essays provided by a college consulting firm, all labeled with expert reviewer scores. Our analysis explores both the predictive performance of different model configurations and the impact of individual trait weaknesses on essay ratings. We find that models incorporating both trait scores and textual content modestly outperform trait-only models, and that the presence of a low score in any single dimension is a strong indicator of reduced overall performance.

By blending LLM-driven evaluation with interpretable traits grounded in real-world practice, this work contributes to the development of more nuanced, transparent, and context-aware AES systems for narrative writing domains.

2 Background

2.1 Related Work

The education industry has traditionally relied on human reviewers to evaluate and provide feedback on student application essays, a process that is both costly and labor-intensive. As a result, there has been significant interest in Automated Essay Scoring (AES) as a potential solution to expedite essay evaluation. Despite inconsistencies in human grading, AES models have demonstrated a strong ability to match human scores [1].

While numerous AES models have been developed, recent works like Ruijie et al. [2] and Xiao et al. [3] highlight the power of Large Language Models (LLMs) for this task, which exhibit notable consistency, generalizability, and explainability. However, while LLMs have made significant strides, their accuracy—particularly within the domain of college application essays—still has room for improvement.

2.2 Limitations with AES and LLMs

Current LLM-based AES systems tend to provide generic, unhelpful feedback. In the context of Common App essays, where factors such as personal voice, emotional vulnerability, and storytelling matter deeply, the scoring rubric must be far more nuanced and critical than for academic writing. This study adopts a more precise set of evaluative criteria, inspired by professional consulting practices.

2.3 Novelties

Our system distinguishes itself through its domain-specific understanding of what makes a compelling Common App essay. Rather than relying solely on generic language features, we introduce and compute five targeted metrics that align with common admissions values and essay evaluation practices:

1. **PST (Problem/Solution/Takeaway) Ratio:** Measures how well an essay balances narrative structure across three key components:
 - **Problem:** Introduces a personal challenge, limitation, or moment of tension.
 - **Solution:** Describes the actions the student took to address or overcome the challenge.

- **Takeaway:** Offers insight or personal growth derived from the experience.

An ideal PST Ratio allocates 20% of the essay to the problem, 60% to the solution, and the remaining 20% for the takeaway.

2. **Specificity:** Measures the vividness and precision of an essay's language. High-scoring essays include concrete details, specific imagery, and nuanced descriptions, while low-scoring essays tend to rely on vague generalities and overused phrases.
3. **Pace:** How effectively the essay varies its sentence lengths to create dynamic storytelling. Essays that exhibit a natural, engaging rhythm with a mix of short, medium, and long sentences must have higher pace scores.
4. **VSPICE:** How well the essay demonstrates six core character traits: **V**ulnerability, **S**elflessness, **P**erseverance, **I**nitiative, **C**uriosity, and **E**xpression. Balanced essays that integrate both emotional and intellectual development across these traits receive higher scores.
5. **Creativity:** How original the essay's content and angle are compared to typical college application essays. Creativity is based on the uniqueness of the story, the perspective, and the style of expression.

These five metrics serve as the foundation for holistic scoring, offering a structured way to assess the depth, clarity, and impact of a student's personal essay.

3 Methodology

3.1 Data Collection

This study uses a dataset provided by ElevatEd, a college consulting firm, comprising approximately 4,000 college application essays submitted by real students. Each entry in the dataset includes:

- A Google Docs link to the original student-authored essay, along with corresponding human feedback.
- A holistic score assigned by expert human reviewers, ranging from <5 to 9.
- A categorical label identifying the type of essay. For the purposes of this study, only essays labeled as “Common App” were included.

All students whose essays were included in the dataset provided their informed consent for their work to be shared and analyzed.

3.2 Data Preprocessing

From the entire dataset, a filtered subset was constructed by selecting only entries corresponding to Common App essays. To isolate the essay content from the accompanying feedback, we developed an algorithm powered by OpenAI’s GPT-4o model to extract only the student-written portion from each Google Document.

Each essay was paired with its holistic score. Because ElevatEd’s rubric includes a range of scoring formats (e.g., “7–8” or <5), we normalized scores into single numeric values:

- Split scores were averaged into a single floating-point value (e.g., “7–8” becomes 7.5).
- Scores marked as <5 —the lowest possible rating—were converted to 4.5 for consistency and modeling purposes.

Entries with missing or incomplete data, such as broken links, absent scores, or revoked consent, were excluded. The final cleaned dataset consisted of 1,177 fully processed essays.

3.3 Prompt Engineering

We employed OpenAI’s GPT-4o model to assign numerical scores to each essay across five dimensions: PST Ratio, Specificity, Pace, VSPICE, and Creativity. Each metric was scored on a scale from 1.0 to 10.0. To ensure reliable and consistent evaluations, we used prompt engineering strategies that provided the model with detailed rubrics and precise definitions for each trait.

For the PST Ratio, the model was instructed to label each sentence in the essay as contributing to either the problem, solution, or takeaway. We then calculated the proportion of words falling under each category and compared the resulting distribution to an ideal narrative structure of 20% problem, 60% solution, and 20% takeaway. A final PST score was assigned based on how closely the essay’s structure aligned with this target distribution.

This scoring framework was applied across the full dataset of 1,177 Common App essays, generating structured trait scores for each submission. These scores form the basis of our subsequent analysis, allowing us to explore trends, correlations, and predictors of strong essay performance.

3.4 Data Training

Our goal was to build a predictive model capable of estimating an essay’s overall rating based on both structural trait scores and the essay’s textual content. We began by training a baseline model using only the five engineered metrics—PST Ratio, Specificity, Pace, VSPICE, and Creativity—as input features. A Random Forest Regressor was selected due to its robustness to feature scale and interpretability. However, the model demonstrated limited predictive performance when relying on these trait scores alone, suggesting that additional information embedded in the essay’s raw text could improve accuracy.

To enhance performance, we expanded our feature space by incorporating representations of the essay text. First, we applied a TF-IDF vectorization strategy, transforming each essay into a fixed-length vector of the 300 most informative words. These features were concatenated with the five trait scores to form a hybrid feature set, which was then used to train a new model. While performance improved modestly, we hypothesized that deeper semantic information could further enhance the model’s ability to capture essay quality.

To that end, we experimented with transformer-based language models, using a sentence-level BERT encoder (specifically, the `all-MiniLM-L6-v2` model) to generate dense embeddings for each essay. These 384-dimensional semantic vectors were combined with the trait scores and used as input to various regressors, including Random Forest and Gradient Boosting models.

4 Results

4.1 Holistic Score Prediction

Our goal was to predict the overall essay rating based on both engineered trait scores and textual content. We began with a baseline linear regression model using only the five extracted metrics (PST Ratio, Specificity, Pace, VSPICE, and Creativity), but this approach showed minimal correlation with the actual human-assigned ratings, as visualized in the correlation matrix (see Figure 4.1).

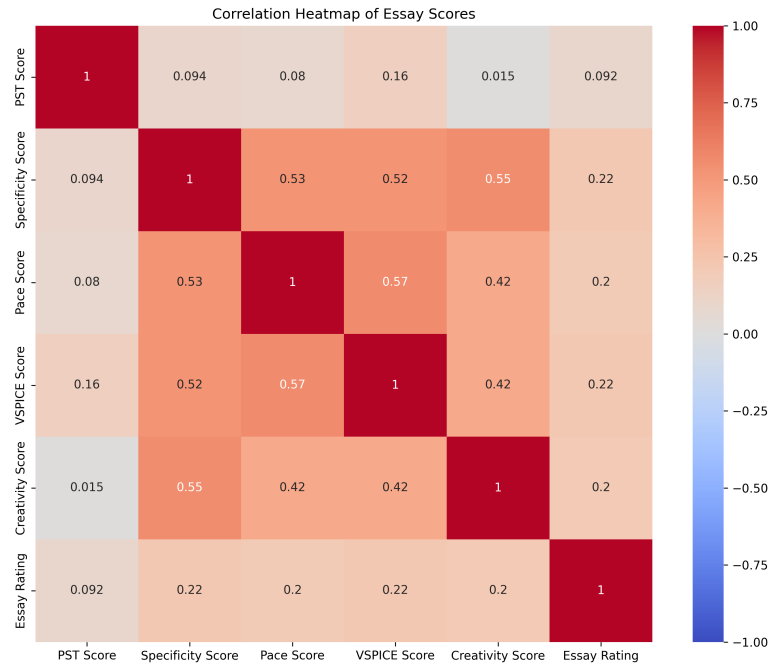


Figure 4.1: Correlation heatmap showing relationships between trait scores and overall essay rating.

The heatmap reveals that all five traits are positively correlated with one another, as well as with the holistic essay rating. While the strength of these correlations is moderate, the pattern suggests that strong performance in one dimension—such as Specificity or VSPICE—is often accompanied by stronger performance in others. This reinforces the idea that high-quality essays tend to excel across multiple dimensions, and that these traits collectively contribute to the perception of overall essay strength.

We then trained a Random Forest Regressor on the same five metrics. This model yielded an R^2 score of **-0.0032** and a mean squared error (MSE) of **1.0038**, indicating that

it performed no better than a naive mean-based predictor. The poor performance suggests that while the extracted metrics are conceptually meaningful, they may not individually or collectively capture enough signal to accurately predict holistic ratings—possibly due to noise and subjectivity in human evaluation.

To address this limitation, we incorporated the full essay text using TF-IDF vectorization. When combined with the trait scores, the Random Forest model achieved improved results: an R^2 score of **0.1554** and an MSE of **0.8451**. This marked improvement indicated that textual features contribute useful predictive information beyond structured trait scores.

Seeking deeper semantic insight, we replaced TF-IDF vectors with dense embeddings from a sentence-level BERT encoder (`all-MiniLM-L6-v2`). Surprisingly, the performance remained essentially unchanged: the model again achieved an R^2 score of **0.1554** and an MSE of **0.8451**. Similarly, experimenting with alternative modeling strategies such as gradient boosting did not yield significant gains.

These results suggest a performance ceiling—likely due to the inherent subjectivity and variability in human scoring. Even with rich textual embeddings and advanced modeling, predicting holistic essay ratings remains a challenging task, potentially requiring larger datasets, more fine-grained annotation, or models fine-tuned specifically for this domain.

4.2 Impact of Low Trait Scores on Overall Essay Rating

We hypothesized that if an essay exhibited a significantly low score in any one of its five core dimensions, it would be more likely to receive a lower overall holistic rating. To test this, we defined a low trait score as any score falling below a threshold of 5.0 and created a binary feature, `has_low_score`, indicating whether an essay had at least one trait below this threshold.

We then compared the average holistic ratings for two groups: essays with no low trait scores and those with at least one. The results showed a clear difference in mean scores: essays with no low scores averaged **5.93**, while essays with at least one low score averaged only **5.49**.

To visualize this, we created a boxplot (Figure 4.2) comparing the distribution of essay ratings across the two groups. Essays with at least one low trait score not only had a lower median but also a lower maximum observed rating, suggesting a performance ceiling effect.

To determine whether this difference was statistically significant, we conducted Welch’s two-sample t -test. The result confirmed a significant difference between the groups ($t = 4.92$, $p < 0.0001$), indicating that the presence of a low score in any single dimension is a strong negative signal for overall essay quality.

These findings support the notion that even one underdeveloped trait—such as vague language, flat pacing, or a lack of vulnerability—can have a disproportionate impact on how readers perceive the strength of a personal statement.

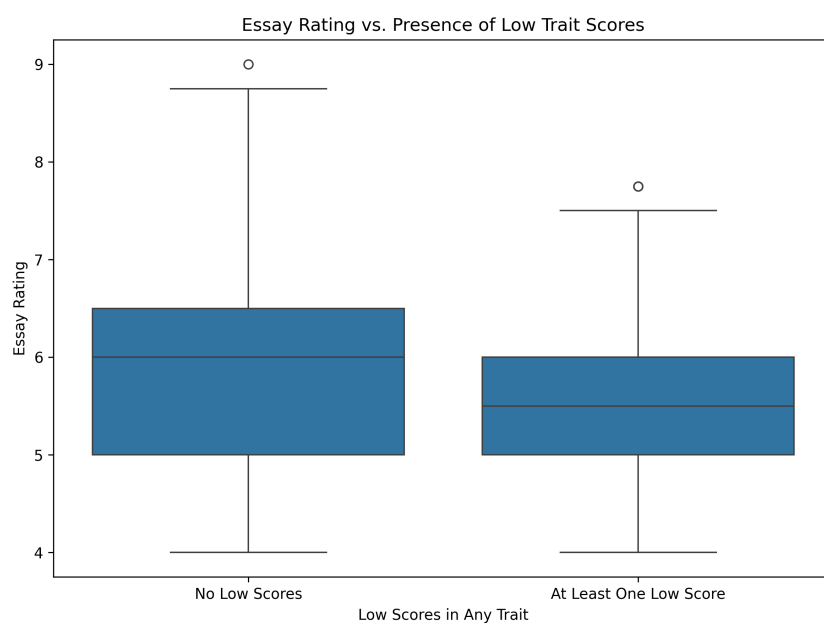


Figure 4.2: Boxplot comparing essay ratings for submissions with and without low trait scores. Essays with at least one trait score below 5.0 tend to receive lower overall ratings, with both median and maximum scores significantly reduced compared to essays without low scores.

5 Conclusions

The findings of this study offer several important insights into the challenges and opportunities of automated essay scoring (AES) in the domain of college application essays.

First, the moderate correlations observed between the engineered trait metrics and overall essay ratings suggest that these traits—while conceptually meaningful—do not fully capture the complexity of human judgment. The fact that high scores in one dimension are often accompanied by high scores in others reinforces the value of a multi-dimensional evaluation framework. However, the limited predictive power of models trained solely on these metrics indicates that further refinement or additional features may be necessary to model holistic judgments more effectively.

Second, the incorporation of raw essay text, particularly through TF-IDF and BERT-based embeddings, resulted in meaningful performance gains. This highlights the importance of capturing semantic content, rhetorical structure, and writing style—factors that are difficult to reduce to a small set of numeric scores. Nevertheless, the performance plateau across modeling approaches suggests diminishing returns without further innovation in either feature extraction or model architecture.

One of the most compelling findings was the clear impact of low trait scores on holistic ratings. The presence of even a single underdeveloped trait (e.g., vague language or lack of emotional depth) was strongly associated with reduced overall scores. This suggests that admissions reviewers may be particularly sensitive to essays that feel unbalanced or incomplete, and that AES systems should pay close attention to identifying and flagging these weaknesses.

Finally, our results underscore the inherent subjectivity and variability in human scoring of personal narratives. Essays may resonate differently with different reviewers, depending on values, lived experience, or expectations. As such, it may be more realistic to view AES as a tool for formative feedback and revision support, rather than as a perfect proxy for human judgment.

6 Future Work

Future work may benefit from larger and more diverse datasets, finer-grained annotation (e.g., reviewer comments or paragraph-level scoring), and model architectures explicitly designed for narrative understanding. In particular, integrating attention-based models that can align specific essay segments with rubric dimensions could offer greater interpretability and fidelity to human scoring logic.

Bibliography

- [1] Austin Pack, Alex Barrett, and Juan Escalante. “Large language models and automated essay scoring of English language learner writing: Insights into validity and reliability”. In: *Computers and Education: Artificial Intelligence* 6.100234 (2024). ISSN: 2666-920X. DOI: 10.1016/j.caeai.2024.100234. URL: <https://www.sciencedirect.com/science/article/pii/S2666920X24000353>.
- [2] Ruijie Liu et al. “A comprehensive analysis of evaluating robustness and generalization ability of models in AES”. In: *Journal of Physics: Conference Series*. Vol. 2813. 1. 2024, p. 012022. DOI: 10.1088/1742-6596/2813/1/012022.
- [3] Changrong Xiao et al. “Human-AI Collaborative Essay Scoring: A Dual-Process Framework with LLMs”. In: *arXiv preprint arXiv:2401.06431* (2024). DOI: 10.48550/arXiv.2401.06431. URL: <https://doi.org/10.48550/arXiv.2401.06431>.